

AY25 PROGRAM ASSESSMENT REPORT

v2026.1

Tracking Data

CIM code (e.g., ME-CM)	BS-CHEM-3
Program Name	Bachelor of Science in Chemistry
Assessment Year	2024-2025
Program Assessment Coordinator	Katie Leight
Year of Current Assessment Plan	2024-2025
No Assessment This Year per Plan	☐
LO(s) Assessed	LO 1c, 1f, and 2
Presented at program authority meeting date: <i>Please attach minutes</i>	10/8/2025 Chemistry Faculty Meeting

Methodology

For each LO and artifact assessed, please describe methodology used, and attach (as supplemental files during report submission) or insert here the rubric used. At minimum this should contain:

- a description of how the artifact is concordant with the LO
- a rubric describing the method used to place individual student in at least 3 assessment categories; {**D** Does not meet expectations, **M** Meets expectations **M**, **P** Proficient}

LO 1c: Students will have foundational technical knowledge. They will be able to critically analyze and respond to chemical questions in relevant areas: Organic chemistry.

The course we will assess this learning objective is CHEM 239 – Organic Chemistry II. The artifact that we will use is the ACS exam that was used for the Final Exam from Spring 2025. The national average on this exam was a 36.7/70. Composite norms from the ACS for the Organic Chemistry exam are included in the attachments.

	Does not meet expectations (D or 1)	Meets Expectations (M or 3)	Proficient (P or 5)
LO 1 c) Foundational Organic Chemistry	Students are unable to answer more that 36.7/70 of foundational organic chemistry questions, and performed below the national average.	Students are able to answer more that 36.7/70 of foundational organic chemistry questions on the ACS exam, performing above the national	Students perform above 80 percentile (46/70) and are able to answer most foundational organic chemistry questions on the ACS exam,

		average but below 80 percentile (46/70) and are missing a couple key concepts	demonstrating a thorough understanding of foundational organic chemistry
--	--	---	--

LO 1f: Students will have foundational technical knowledge. They will be able to critically analyze and respond to chemical questions in relevant areas. For students in the concentrations, they will additionally have foundational technical knowledge in: Environmental chemistry (Environmental Chemistry Concentration).

The course we will use to assess this learning objective is CHEM 475 – Environmental Chemistry. The artifact that we will use is the Final Exam from Fall 2024.

	Does not meet expectations (D or 1)	Meets Expectations (M or 3)	Proficient (P or 5)
LO 1 f) Foundational Environmental Chemistry	Students are unable to answer more than 50% of foundational Environmental Chemistry	Students are able to answer more than 50% of foundational organic chemistry but are missing a couple key concepts	Students are able to answer over 80% of foundational environmental chemistry questions, demonstrating a thorough understanding of foundational environmental chemistry

LO 2: Students will have problem-solving skills. They will be able to: a) Explain the rationale behind chemistry experiments b) Demonstrate technical skills to carry out experiments in the laboratory or computer and to record results c) Analyze data using qualitative and quantitative methods, including computer software, and apply critical reasoning to draw appropriate conclusions

The courses we will use to assess this learning objective are Chem 344 – Physical Chemistry 2 (Lab) from both Fall 2024 and Spring 2025 semesters and Chem 416 – Advanced Chemistry Lab from Spring 2025 semester. For Chem 344, the rubric below was used.

	Does not meet expectations (D or 1)	Meets Expectations (M or 3)	Proficient (P or 5)
LO 2) Problem Solving Skills	Students are unable to explain the rationale	Students are able to explain the rationale	Students are more than capable of explaining

	behind the experiment, demonstrate technical skills to carry out the experiment, or analyze the data, receiving below a 50% on the lab report.	behind the experiment, demonstrate technical skills to carry out the experiment, and analyze the data, but are missing a couple key concepts, receiving below a 80% on the lab report	the rationale behind the experiment, demonstrating technical skills to carry out the experiment, and analyzing the data, receiving above a 80% on the lab report
--	---	--	---

Results

Artifact samples

Attach artifact samples. You can paste them in here if feasible, attach them to the end of this report, or submit them as supplemental files when submitting your reports. For data sets with ≤ 10 students, provide 1 in each category, D,M,P, (unless the category is empty, if so state that). For programs with ≤ 20 , provide 2. For other programs provide 3. Please redact any personally identifiable information, PII (which includes names and A#s). Please convert your documents to PDFs.

Please use the following naming convention for your artifact files: **AY25_CIM-CODE_Name of Artifact.pdf**.

LO 1c: (Actual ACS exam is not included as it would violate the security pledge with the American Chemical Society) The table below shows the chemistry majors' scores on the ACS exam who were enrolled in Chem 239 during the Spring 2025 semester.

Student	ACS Final Score	Percentile (national)	D/M/P
A	43/70	72	M
B	60/70	97	P
C	29/70	28	D
D	39/70	61	M
E	31/70	35	D

LO 1f: Scanned copies of the chemistry majors' final exams are included as a supplemental file. The table below shows their scores on the final exam of the chemistry majors who were enrolled in Chem 475 during the Fall 2024 semester.

Student	Final Exam Score	D/M/P
A	31/100	D
B	46/100	D
C	27/100	D
D	50/100	M

LO 2: Scanned copies of the chemistry majors' Lab Reports for the Conjugated Dyes lab from Chem 344 both fall 2024 and spring 2025 semesters are included as a supplemental file.

Student	Percentage	D/M/P
A	100% (30/30)	P

B	99	P
C	93	P

Scanned copies of the chemistry majors' Lab reports as well as their feedback for the Lab 2: Synthesis and Doping of Iron Oxide Nanoparticles from Chem 416 in Spring 2025 are included as a supplemental file.

student	Grade	D/M/P
A	95/100	P
B	92/100	P
C	94/100	P
D	93/100	P
E	94/100	P
F	94/100	P
G	92/100	P
H	95/100	P
I	90/100	P
J	90/100	P
K	95/100	P

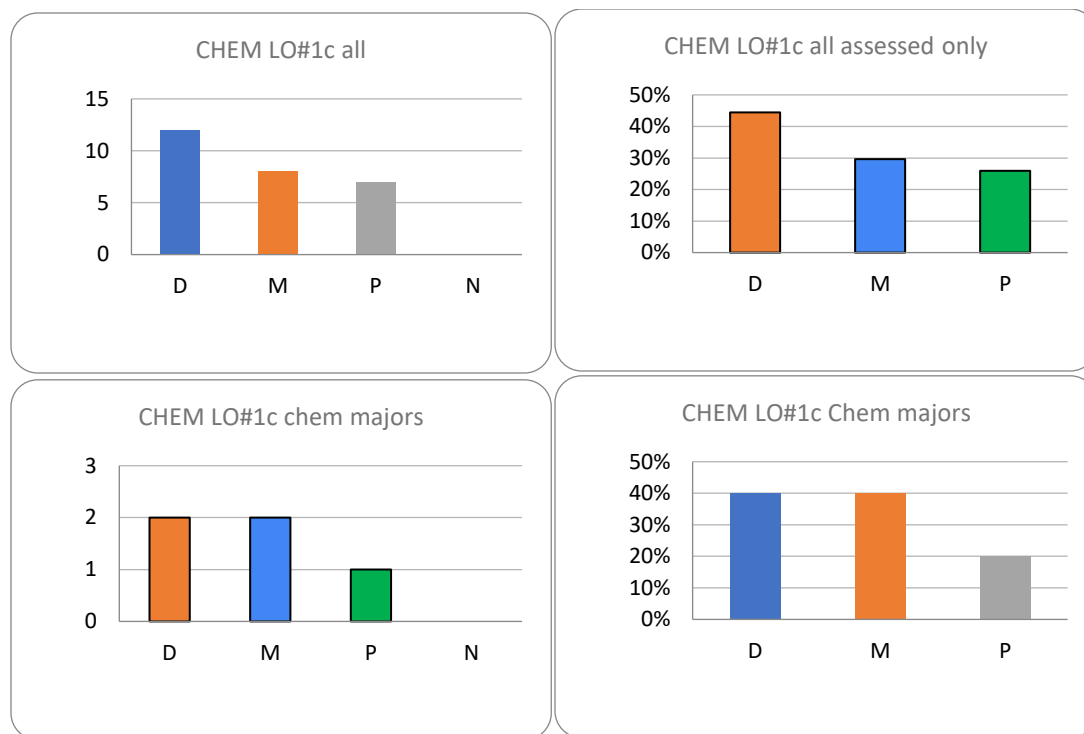
Data table

For each LO and provide a table pasted in from the [provided template](#) placing each student into the assessment categories (minimally {D, M, P}). Please redact any personally identifiable information, PII (including names or A#s) from this report. Provide a summary graph from the template presenting this distribution. You should also present additional analysis as deemed appropriate by the results seen, and the needs of the program.

LO 1c:

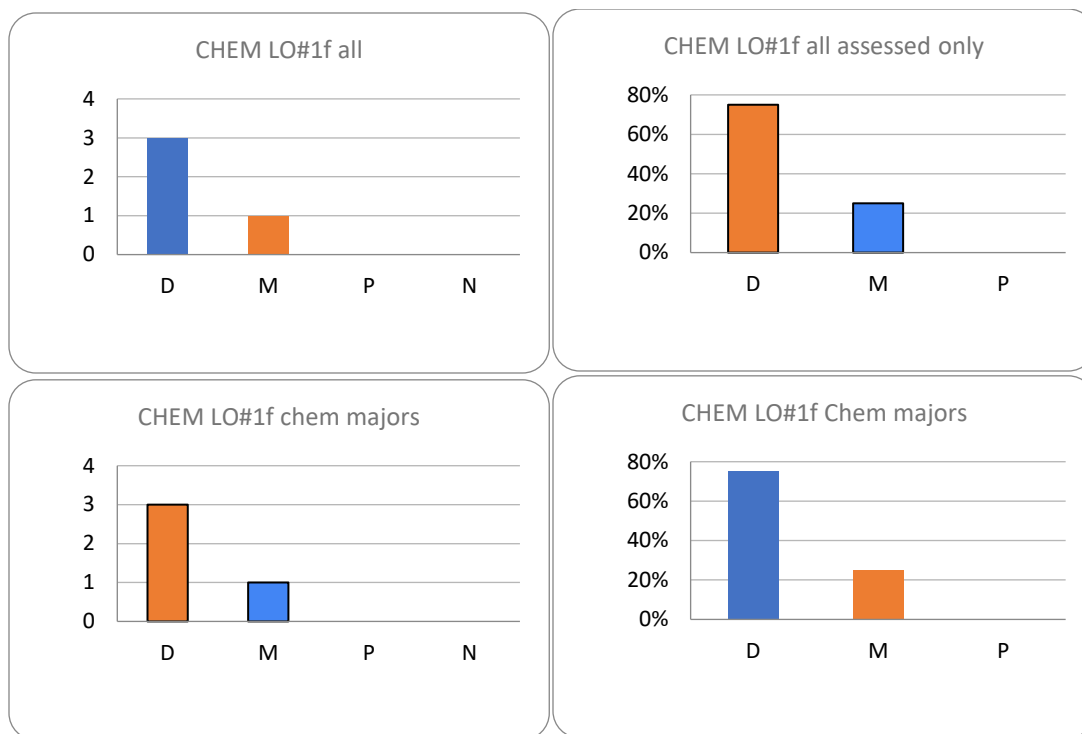
student	major	D/M/P	comment for N
student 1	CHEM	M	
student 2	CHEM	P	
student 3	CHEM	D	

student 4	CHEM	M	
student 5	CHEM	D	
student 6	NON CHEM	M	
student 7	NON CHEM	D	
student 8	NON CHEM	P	
student 9	NON CHEM	D	
student 10	NON CHEM	D	
student 11	NON CHEM	D	
student 12	NON CHEM	D	
student 13	NON CHEM	P	
student 14	NON CHEM	D	
student 15	NON CHEM	D	
student 16	NON CHEM	D	
student 17	NON CHEM	P	
student 18	NON CHEM	M	
student 19	NON CHEM	P	
student 20	NON CHEM	D	
student 21	NON CHEM	M	
student 22	NON CHEM	P	
student 23	NON CHEM	M	
student 24	NON CHEM	P	
student 25	NON CHEM	M	
student 26	NON CHEM	M	
student 27	NON CHEM	D	



LO 1f:

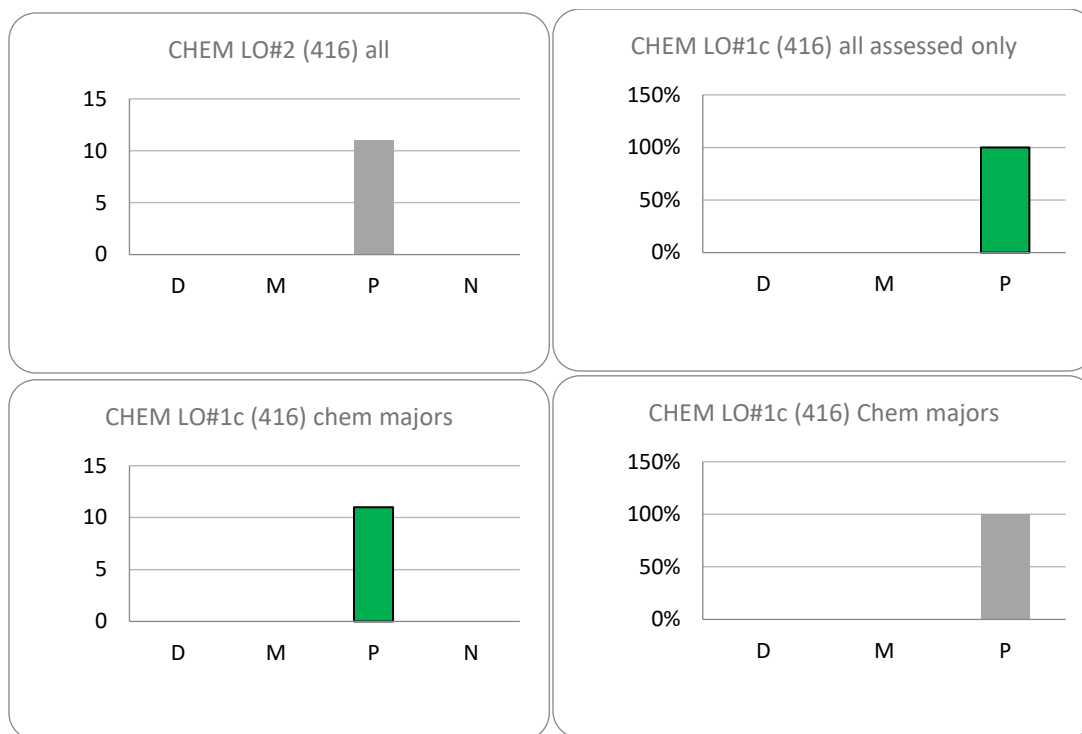
student	major	D/M/P
student 1	CHEM	D
student 2	CHEM	D
student 3	CHEM	D
student 4	CHEM	M



LO 2:

Chem 416

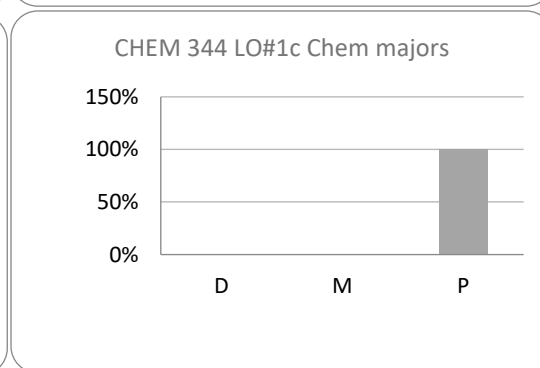
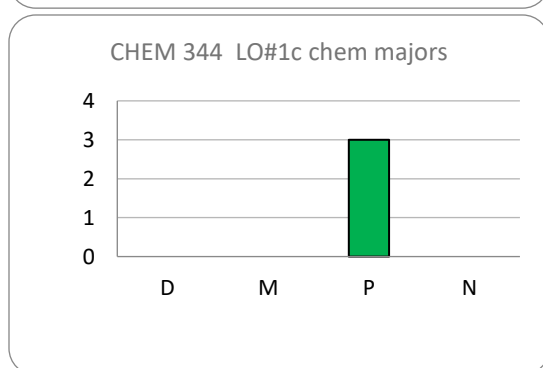
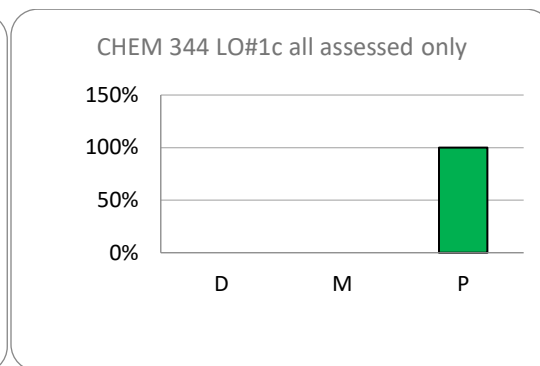
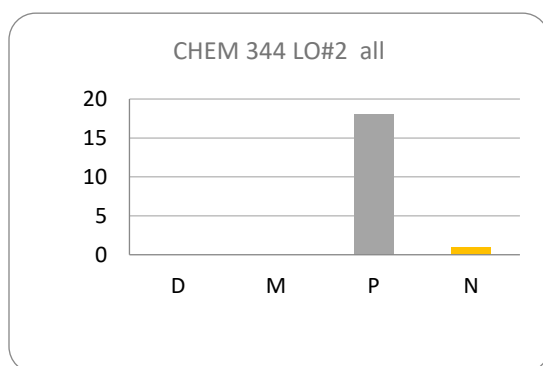
student	major	D/M/P
student 1	CHEM	P
student 2	CHEM	P
student 3	CHEM	P
student 4	CHEM	P
student 5	CHEM	P
student 6	CHEM	P
student 7	CHEM	P
student 8	CHEM	P
student 9	CHEM	P
student 10	CHEM	P
student 11	CHEM	P



Chem 344 (lab)

student	major	D/M/P	comment for N
student 1	CHEM	P	
student 2	CHEM	P	
student 3	CHEM	P	
student 4	NON CHEM	P	
student 5	NON CHEM	P	
student 6	NON CHEM	N	student did not turn in lab report
student 7	NON CHEM	P	
student 8	NON CHEM	P	
student 9	NON CHEM	P	
student 10	NON CHEM	P	
student 11	NON CHEM	P	
student 12	NON CHEM	P	
student 13	NON CHEM	P	

student 14	NON CHEM	P	
student 15	NON CHEM	P	
student 16	NON CHEM	P	
student 17	NON CHEM	P	
student 18	NON CHEM	P	
student 19	NON CHEM	P	



Description of findings

Summarize the discussion that occurred within the program authority when this was presented. Provide a short narrative interpreting the findings. Are there any areas of concern?

Students taking Chem 475 (Environmental Chemistry) seem to be lacking in foundational technical knowledge in Environmental Chemistry.

Continuous Improvement, CI, plans

The purpose of assessment is to improve the quality of our educational programs. The most important part of the assessment cycle is how we use the data to improve our programs. Examine the data provided above describing student achievement, consider the root cause of any deficits, and if areas for possible improvement emerge, describe plans to achieve this improvement.

Prior CI plan impact

Also, please refer to the previous assessment report for the LOs assessed. Were any CI plans recommended? Were they implemented? Did the plans work (were improvements seen in this cycle)?

☒ no previous CI plans available

This cycle's CI plan

Describe any CI plans developed from this cycle's data, with

- Specific curricular or instructional changes in response to this assessment data,
- A timeline for implementation,
- As well a description of who will participate in and manage these plans

☐ After consideration and presentation to the program authority, no CI plans are recommended.

The Chemistry Curriculum Committee will meet within the next academic year to discuss Environmental Chemistry Course Curriculum (LO 1f).